

A Technical Introduction to Diffusion model

$$q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$

Section 1

$$\alpha_t = (1 - \beta_1) \prod_{i=1}^{t-1} (1 - \beta_i) = e^{\sum_{i=1}^{t-1} \ln(1 - \beta_i)} \rightarrow e^{-\int_0^t \beta(t) dt} \quad \{\displaystyle \bar{\alpha}_t = (1 - \beta_1) \cdots (1 - \beta_t) = e^{\sum_{i=1}^t \ln(1 - \beta_i)} \}$$

Section 2

$$\nabla_x \ln p_\gamma(x|y) = (1 - \gamma) \nabla_x \ln p(x) + \gamma \nabla_x \ln p(x|y) \quad \{\displaystyle \nabla_x \ln p_\gamma(x|y) = (1 - \gamma) \nabla_x \ln p(x) + \gamma \nabla_x \ln p(x|y)\}$$

Section 3

Sample the previous data $x_{t-1} \sim \mathcal{N}(\mu_t(x_t, x_0), \sigma_t^2)$ $\{\displaystyle x_{t-1} \sim \mathcal{N}(\mu_t(x_t, x_0), \sigma_t^2)\}$

Section 4

$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1} | \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$ $\{\displaystyle p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1} | \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))\}$
The goal now is to learn the parameters θ $\{\displaystyle \theta\}$ such that $p_\theta(x_0)$ $\{\displaystyle p_\theta(x_0)\}$ is as close to $q(x_0)$ $\{\displaystyle q(x_0)\}$ as possible. To do that, we use maximum likelihood estimation with variational inference.

Section 5

In DDPM, the sequence of numbers $0 = \sigma_0 < \sigma_1 < \dots < \sigma_T < 1$ $\{\displaystyle 0 = \sigma_0 < \sigma_1 < \dots < \sigma_T < 1\}$ is called a (discrete time) noise schedule. In general, consider a strictly increasing monotonic function σ $\{\displaystyle \sigma\}$ of type $\mathbb{R} \rightarrow (0, 1)$ $\{\displaystyle \mathbb{R} \rightarrow (0, 1)\}$, such as the sigmoid function. In that case, a noise schedule is a sequence of real numbers $\lambda_1 < \lambda_2 < \dots < \lambda_T$ $\{\displaystyle \lambda_1 < \lambda_2 < \dots < \lambda_T\}$. It then defines a sequence of noises $\sigma_t := \sigma(\lambda_t)$ $\{\displaystyle \sigma_t := \sigma(\lambda_t)\}$, which then derives the other quantities $\beta_t = 1 - 1 - \sigma_t^2 = 1 - \sigma_{t-1}^2$ $\{\displaystyle \beta_t = 1 - \sigma_t^2 = 1 - \sigma_{t-1}^2\}$. In order to use arbitrary noise schedules, instead of training a noise prediction model $\epsilon_\theta(x_t, t)$ $\{\displaystyle \epsilon_\theta(x_t, t)\}$, one trains $\epsilon_\theta(x_t, \sigma_t)$ $\{\displaystyle \epsilon_\theta(x_t, \sigma_t)\}$. Similarly, for the noise conditional score network, instead of training $f_\theta(x_t, t)$

$f_\theta(x_t, t)$, one trains $f_\theta(x_t, \sigma_t)$ $\{\displaystyle f_\theta(x_t, \sigma_t)\}$.

Section 6

where $\nabla_x \ln p(x)$ $\{\displaystyle \nabla_x \ln p(x)\}$ is the score function, trained as previously described, and $\nabla_x \ln p(y|x)$ $\{\displaystyle \nabla_x \ln p(y|x)\}$ is found by using a differentiable image classifier. During the diffusion process, we need to condition on the time, giving $\nabla_x \ln p(x_t | y, t) = \nabla_x \ln p(y | x_t, t) + \nabla_x \ln p(x_t | t)$ $\{\displaystyle \nabla_x \ln p(x_t | y, t) = \nabla_x \ln p(y | x_t, t) + \nabla_x \ln p(x_t | t)\}$. Although, usually the classifier model does not depend on time, in which case $p(y | x_t, t) = p(y | x_t)$ $\{\displaystyle p(y | x_t, t) = p(y | x_t)\}$. Classifier guidance is defined for the gradient of score function, thus for score-based diffusion network, but as previously noted, score-based diffusion models are equivalent to denoising models by $\epsilon_\theta(x_t, t) = -\sigma_t \nabla_x \ln p(x_t | t)$ $\{\displaystyle \epsilon_\theta(x_t, t) = -\sigma_t \nabla_x \ln p(x_t | t)\}$, and similarly, $\epsilon_\theta(x_t, y, t) = -\sigma_t \nabla_x \ln p(x_t | y, t)$ $\{\displaystyle \epsilon_\theta(x_t, y, t) = -\sigma_t \nabla_x \ln p(x_t | y, t)\}$. Therefore, classifier guidance works for denoising diffusion as well, using the modified noise prediction: $\epsilon_\theta(x_t, y, t) = \epsilon_\theta(x_t, t) - \sigma_t \nabla_x \ln p(y | x_t, t)$ $\{\displaystyle \epsilon_\theta(x_t, y, t) = \epsilon_\theta(x_t, t) - \sigma_t \nabla_x \ln p(y | x_t, t)\}$ classifier guidance $\{\displaystyle \epsilon_\theta(x_t, y, t) = \epsilon_\theta(x_t, t) - \sigma_t \nabla_x \ln p(y | x_t, t)\}$

Section 7

Let the score function be $s(x) := \nabla_x \ln q(x)$ $\{\displaystyle s(x) := \nabla_x \ln q(x)\}$; then consider what we can do with $s(x)$ $\{\displaystyle s(x)\}$. As it turns out, $s(x)$ $\{\displaystyle s(x)\}$ allows us to sample from $q(x)$ $\{\displaystyle q(x)\}$ using thermodynamics. Specifically, if we have a potential energy function $U(x) = -\ln q(x)$ $\{\displaystyle U(x) = -\ln q(x)\}$, and a lot of particles in the potential well, then the distribution at thermodynamic equilibrium is the Boltzmann distribution $q_U(x) \propto e^{-U(x) / k_B T} = q(x)^{1 / k_B T}$ $\{\displaystyle q_U(x) \propto e^{-U(x) / k_B T} = q(x)^{1 / k_B T}\}$. At temperature $k_B T = 1$ $\{\displaystyle k_B T = 1\}$, the Boltzmann distribution is exactly $q(x)$ $\{\displaystyle q(x)\}$. Therefore, to model $q(x)$ $\{\displaystyle q(x)\}$, we may start with a particle sampled at any convenient distribution (such as the standard Gaussian distribution), then simulate the motion of the particle forwards according to the Langevin equation

Section 8

where $\gamma > 0$ $\{\displaystyle \gamma > 0\}$ is interpretable as inverse temperature. In the context of diffusion models, it is usually called the guidance scale. A high γ $\{\displaystyle \gamma\}$ would force the model to sample from a distribution concentrated around $\arg \max_x p(y | x)$

$\arg \max_x p(y|x)$. This sometimes improves quality of generated images. This gives a modification to the previous equation: $\nabla_x \ln p_\beta(x|y) = \nabla_x \ln p_\beta(x) + \gamma \nabla_x \ln p_\beta(y|x)$ $\nabla_x \ln p_\beta(x|y) = \nabla_x \ln p_\beta(x) + \gamma \nabla_x \ln p_\beta(y|x)$ For denoising models, it corresponds to $\epsilon_\theta(x_t, y, t) = \epsilon_\theta(x_t, t) - \gamma \sigma_t \nabla_x \ln p_\theta(y|x_t, t)$ $\epsilon_\theta(x_t, y, t) = \epsilon_\theta(x_t, t) - \gamma \sigma_t \nabla_x \ln p_\theta(y|x_t, t)$